

Data Acquisition

06

Hey Mirai, how does Google Maps always know my accurate location?

It uses the sensors on your mobile device to collect your information.

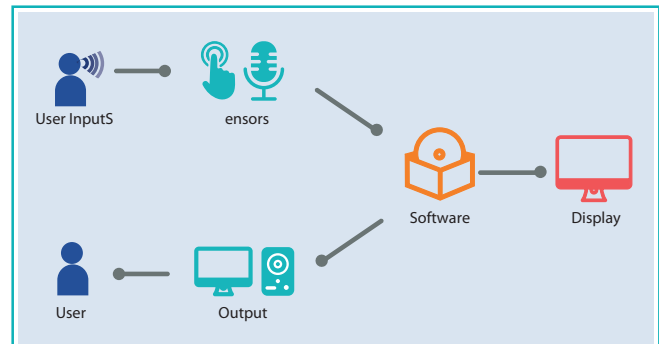


All digital devices need data to operate. The data is collected with the help of sensors. The process of data collection is called data acquisition.

Data Acquisition

Data acquisition is the process that helps devices collect data (numbers, words, images, etc.) from the world and convert it into a digital format. Data needs to be converted into digital form so that it can be analysed in a software and stored in a computer.

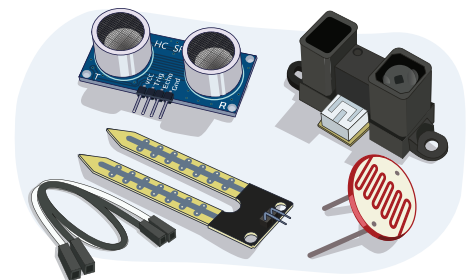
It is collected using various sensors.



Sensors

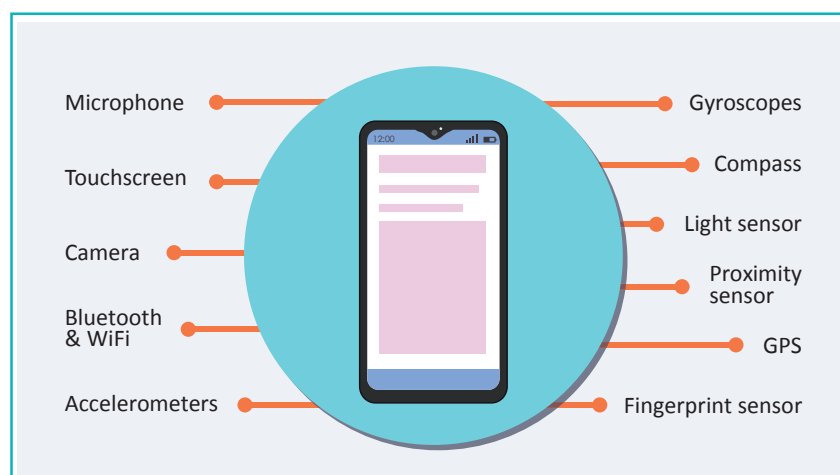
Sensors are electronic devices that detect and measure changes in the nearby environment and send that data to the operating system or processor. They convert physical inputs into electronic signals for the processor or operating system to process.

Physical inputs could be light, heat, motion, moisture, pressure, or any other environmental factor. The output from the sensors is generally an electronic signal that is converted to human-readable value or transmitted electronically to other devices for reading or further processing.



Sensors in a smartphone

Like many other digital devices, smartphones have dozens of sensors built in as per user requirements.



6. Data Acquisition

Some available sensors in smartphones are:

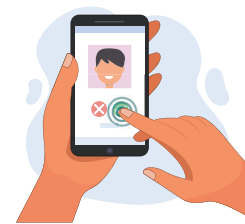
- a. Microphone:** Microphones convert sound signals into electrical signals that help store or transmit data to other smartphones.

For example: Microphones help sound reach other smartphones while calling.



- b. Touchscreen:** The touchscreen receives user input via the touch of a finger and converts it into electrical signals.

For example: When we tap the green icon on the screen, data is transferred to the processor to answer the call.



- c. Digital Camera:** A digital camera captures and records millions of tiny lights in its surroundings and transfers data to the processor in the form of an electrical signal.

For example: A digital camera shutters and captures the light in the area where the camera is pointed at and transforms it into an image.



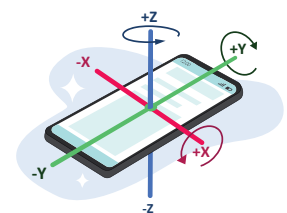
- d. Accelerometer:** An accelerometer measures the linear motion of a smartphone based on vibrations and displays it to the user.

For example: An accelerometer records the forward and backward movement while walking to calculate the total number of steps.



- e. Gyroscope:** A gyroscope helps measure the rotational motion of a smartphone.

For example: A gyroscope tracks the rotation of a phone that enables the screen to auto rotate.



Bytes

The **accelerometer** and **gyroscope** are motion sensors. Motion sensors are useful to monitor device's movements such as tilt, shake, rotation, or swing.

Exercise

Fill in the blanks

- 1 The process of acquiring data from the world around us and converting it into digital form is called _____.
- 2 An electronic device that converts physical inputs from the environment into electronic signals is called a _____.

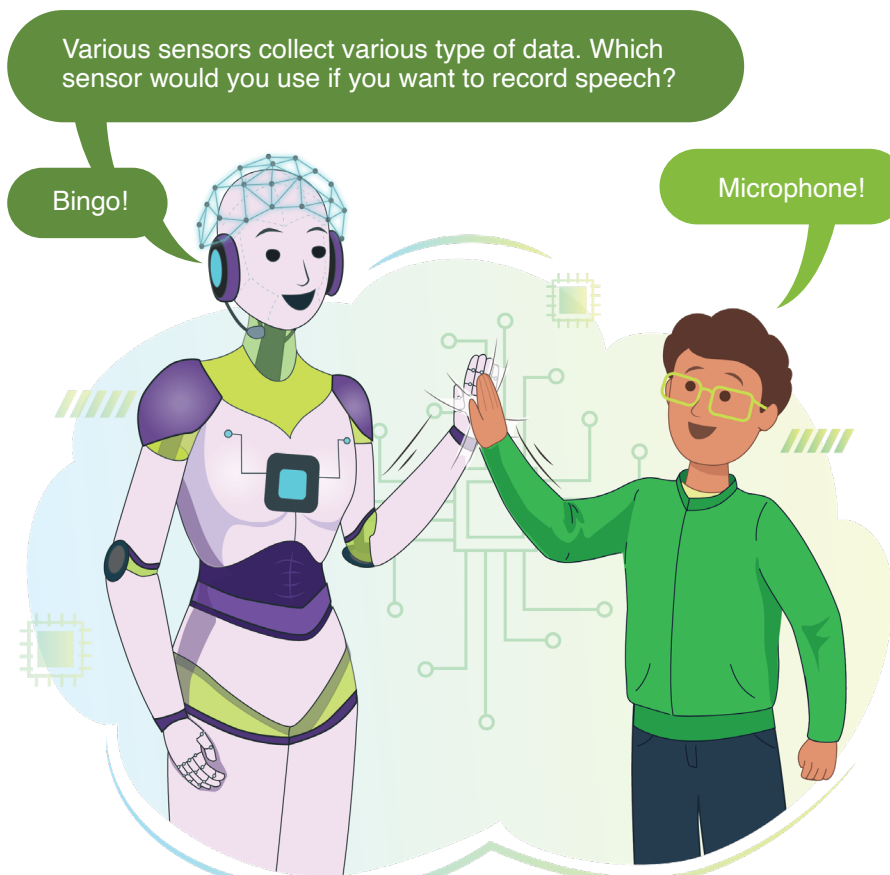
State whether True or False

- 3 Smartphones use sensors to get data from the environment and process it as per the user's requirement.



- 4 An accelerometer measures the linear motion of a smartphone based on vibration.





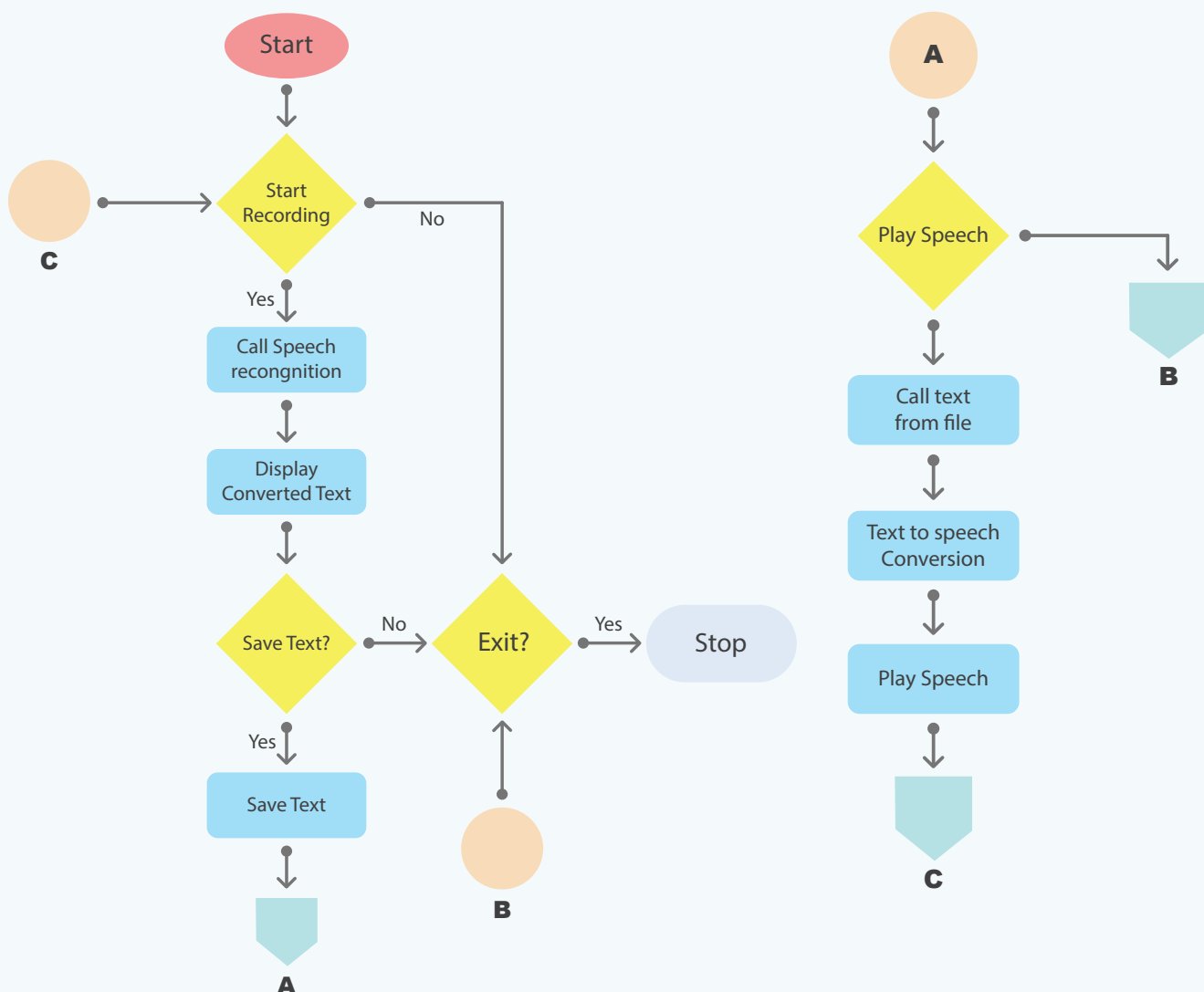
Activity 1

Create an application to record a speech and convert it into text

1 Create a flowchart

Identify the requirements of the application and draw a flowchart to represent them.

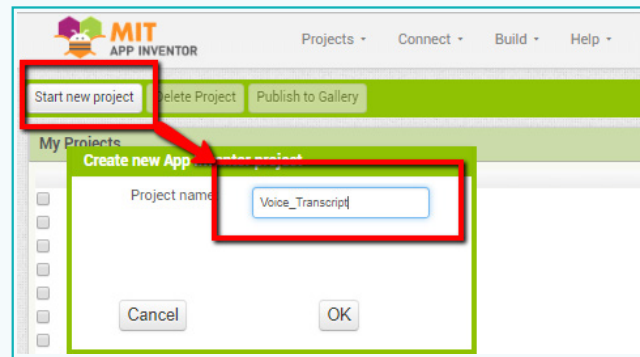
When the application is launched, it will check for the buttons that have and have not been clicked. If Start Recording is clicked, it will recognise the speech and convert it into text and save the text. It also examines if the Play Speech button is clicked or not. If it is clicked, it will call the text of the saved file and convert it into speech.



Activity 1

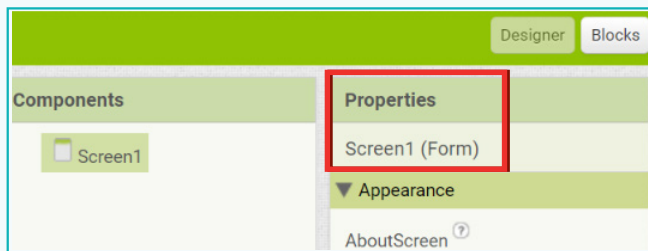
2 Create a new project

- Open MIT App Inventor and create a new project named “Voice Transcript”.



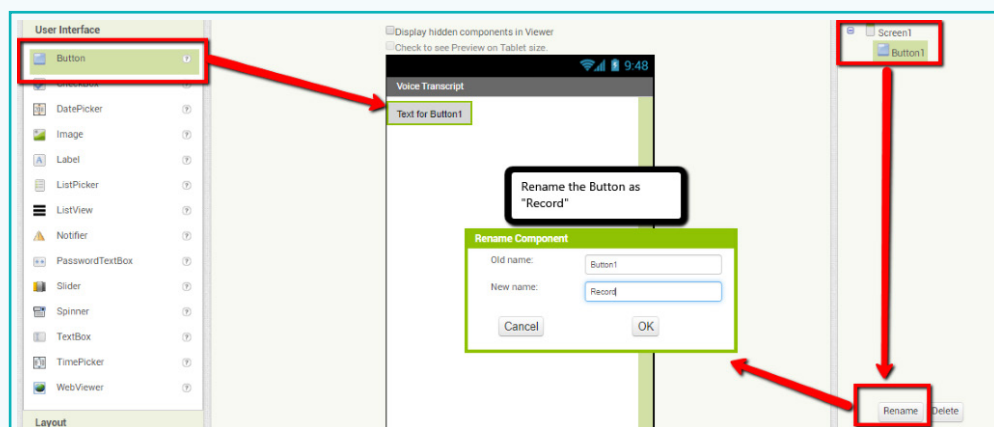
3 UI Design

- Change the screen title to “Voice Transcript” through the Properties section.
- Screen 1 >> Properties >> Title “Voice Transcript”.



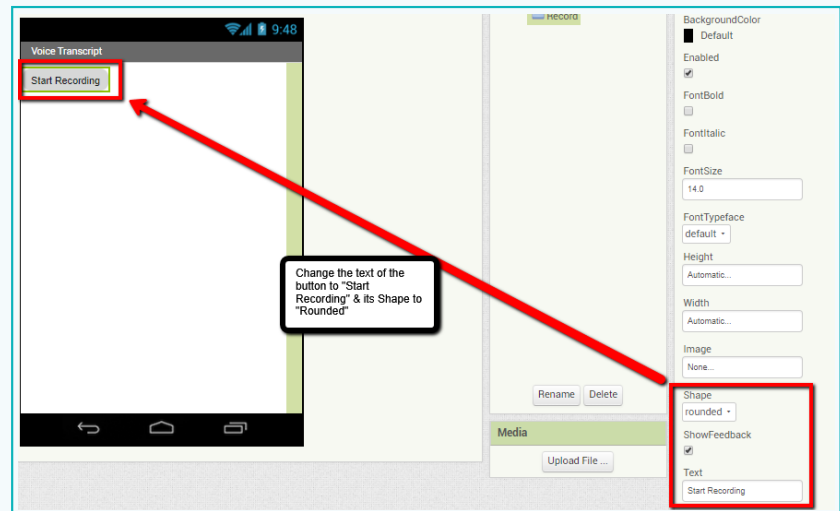
We need a button to start the voice recording.

- Drag a button from the Blocks section & place it in the Viewer section. This will be used to start recording. We will rename it “Record”.
- User Interface >> Button>> Rename as Record

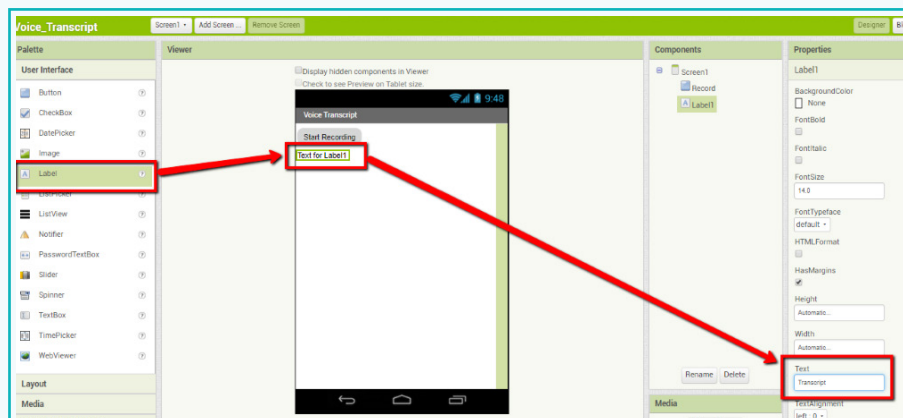


Activity 1

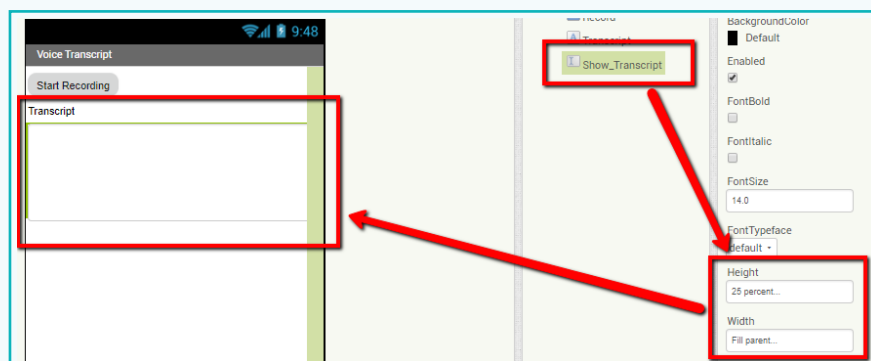
- e. From button properties, change the button text to “Start Recording”.



- f. Place a label and rename it “Transcript”. We will also change its text to say “Transcript”.



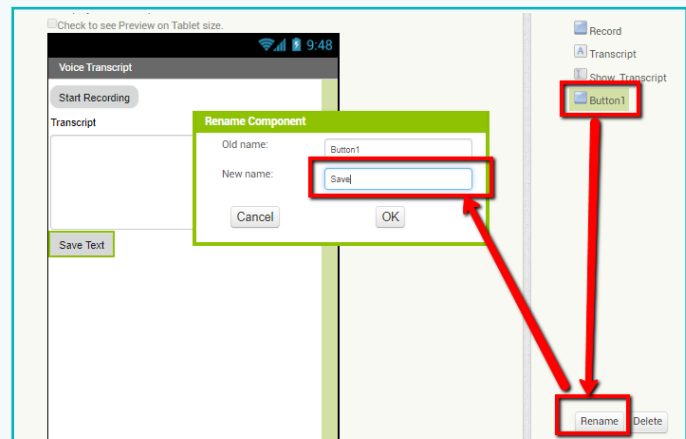
- g. To display the transcript text, place a text box from the block section on the the viewer section. This will display the transcribed text. Rename the component to “Show transcript”.
- h. Resize the text box’s height to “25 percent” & width to “Fill parent”.



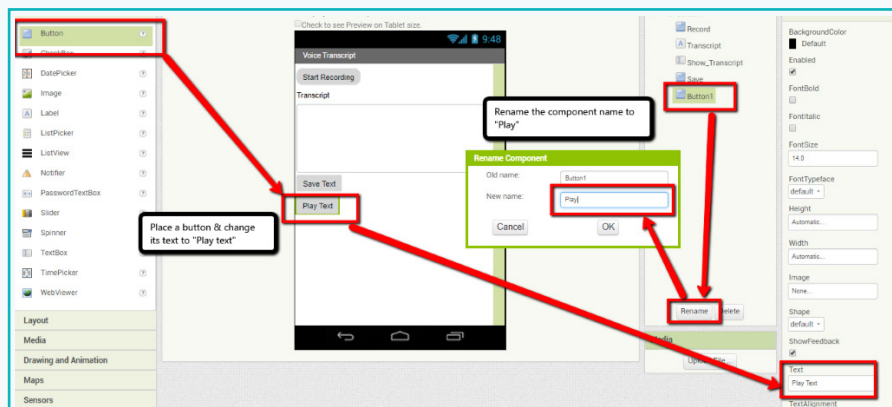
Activity 1

To save the transcript in a text file, place a button to perform the save operation.

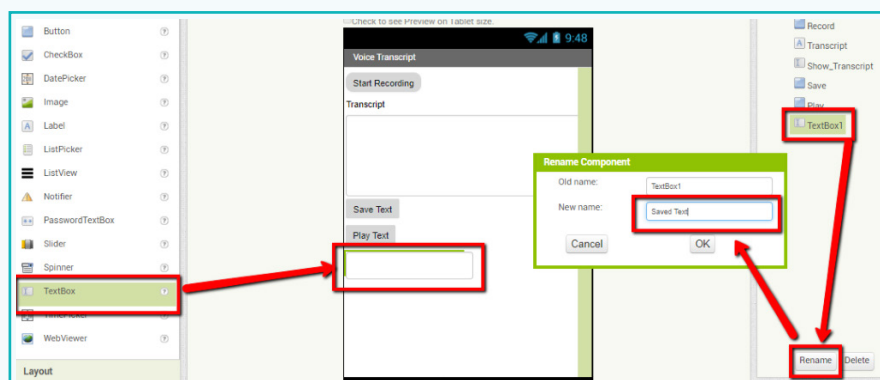
- i. User interface>> Button >> Rename as Save.
- j. Change the button text to “Save Text”.



- k. Place one more button on the viewer screen to call & play the saved text. Change its text to “Play Text” rename the component as “Play”.

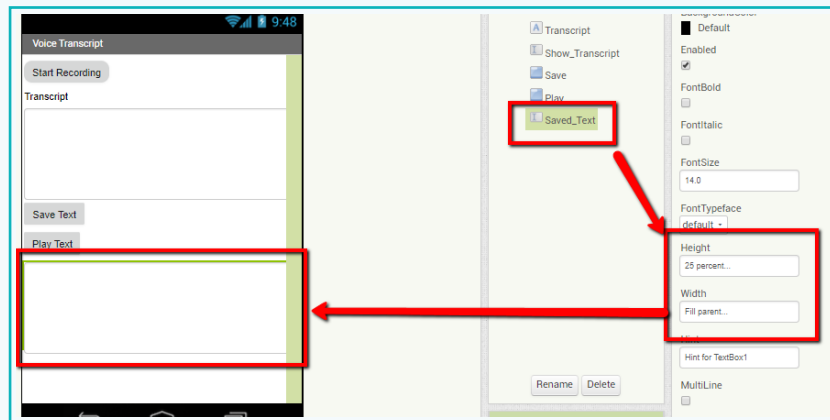


- l. Place a text box from the block section on the viewer section to display the saved text. Rename the component as “Saved text”.

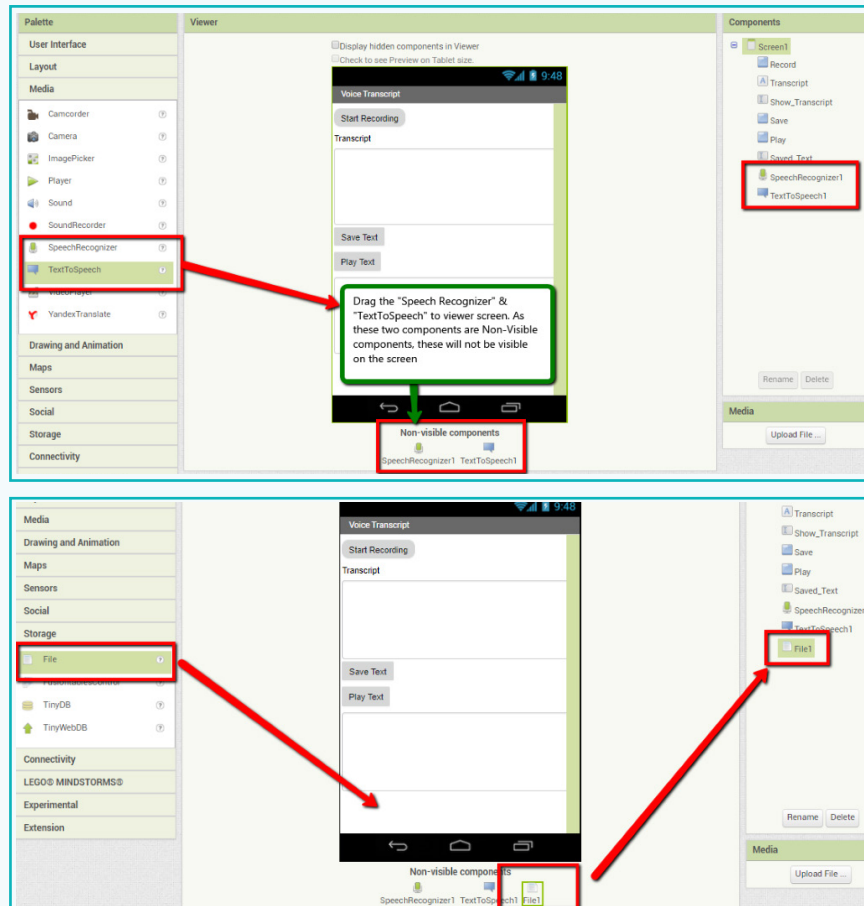


Activity 1

m. User interface>> Button >> Rename as Save.



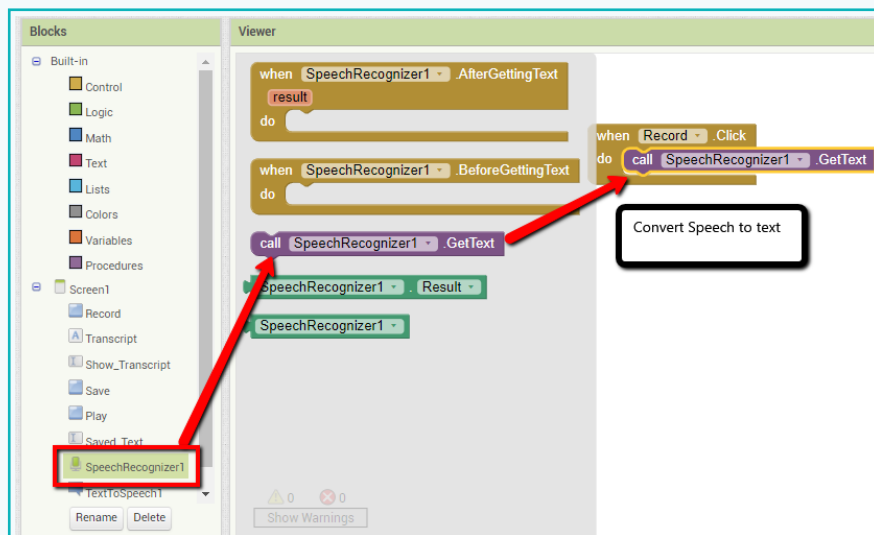
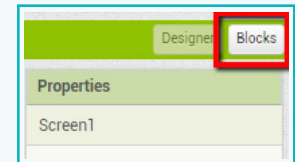
n. Drag the non-visible components “Speech Recognizer”, “TextToSpeech” (from the Media palette) & “File” (from the Storage palette) onto the viewer screen.



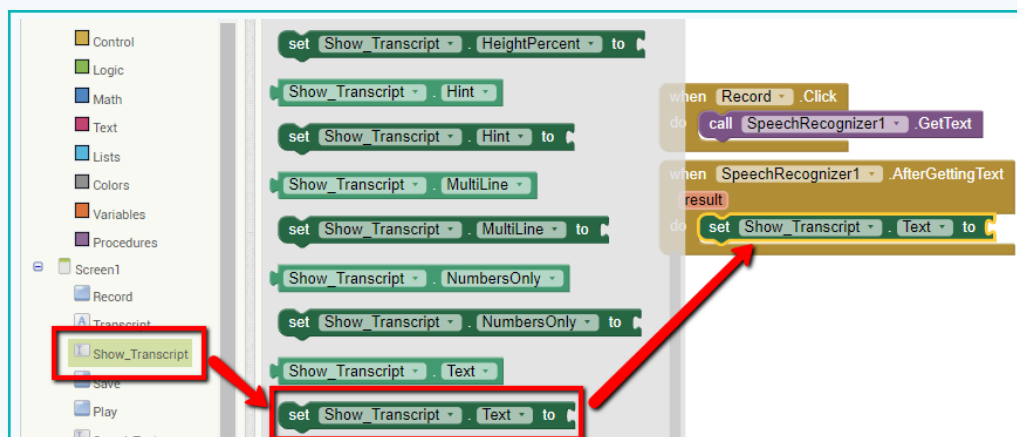
Activity 1

4 Switch to the block screen

- Click “Blocks” to switch to the block Screen and program the functionality of the components.
- From the blocks section, place “when_Record.Click” of the record component on the viewer section.
- To call the speech recognizer to record and transcript a voice, connect “call_SpeechRecognizer.GetText” of SpeechRecognizer1 under “when_Record.Click”.

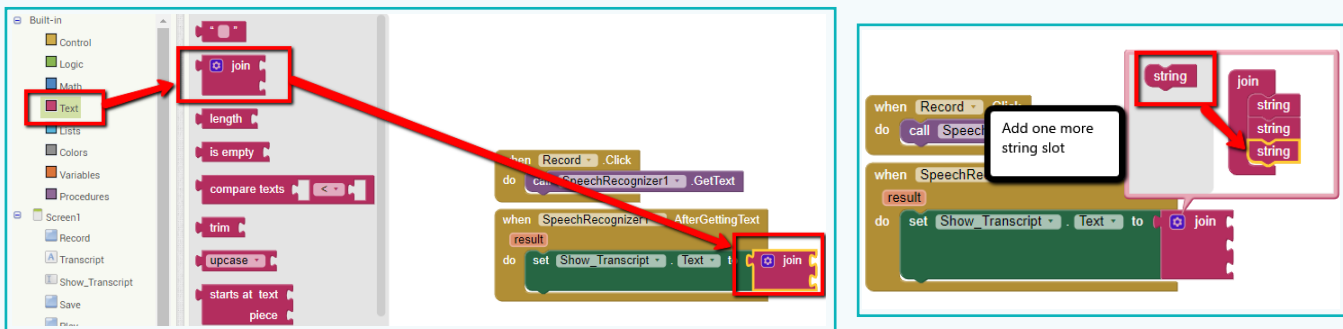


- Place “when_SpeechRecognizer1.AfterGettingText” of SpeechRecognizer1 on the viewer section.
- To display the transcript text, connect “set_Show_Transcript.Text_to” of the transcript component under “when_SpeechRecognizer1.AfterGettingText”.

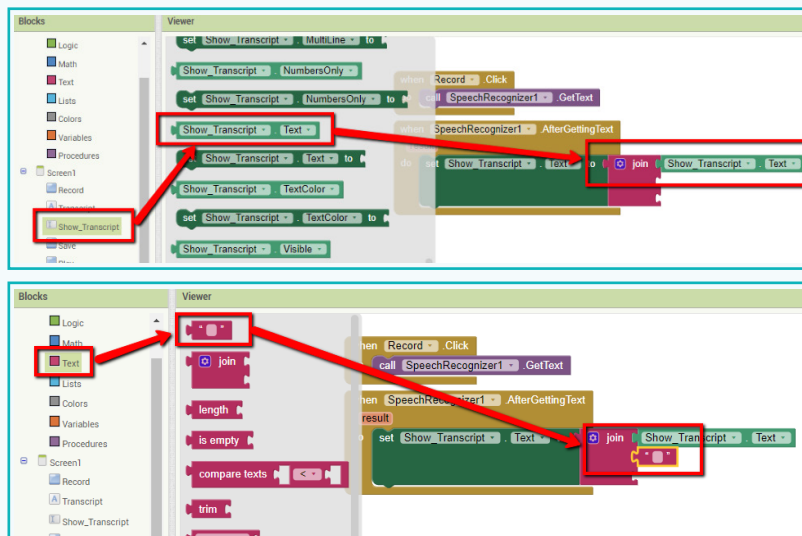


Activity 1

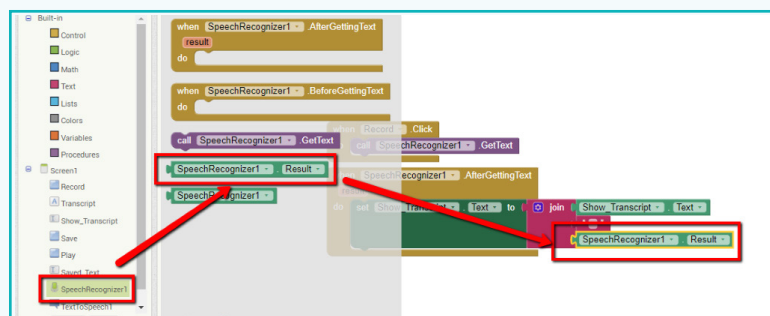
- f. To set the text to display continuously as the voice is recorded, connect “join” from “Text” built-in blocks to “set_Show_Transcript.Text_to”. Click on the settings icon on the “join” block and add another slot for a text constant.



- g. Connect “Show_Transcript.Text” from the “Show_transcript” component to the first slot of “join” block. Connect a text constant to the second slot of “join” block.

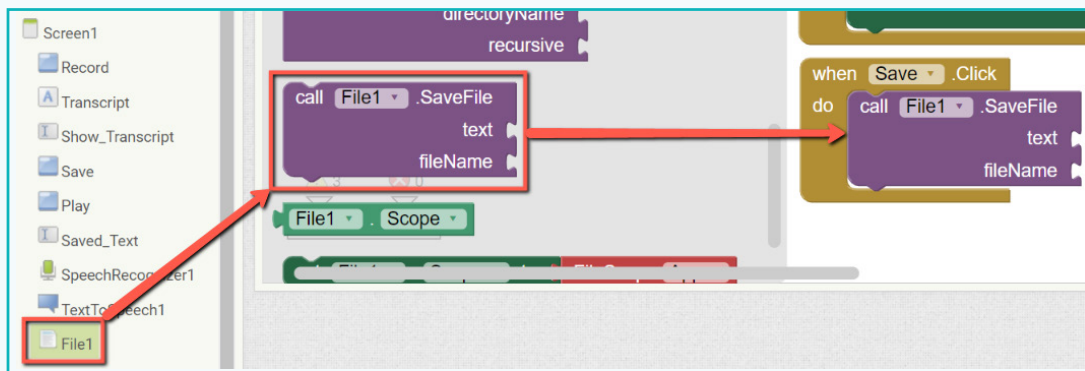


- h. Connect “SpeechRecognizer.Result” from “SpeechRecognizer1” to the third slot of “join” block.

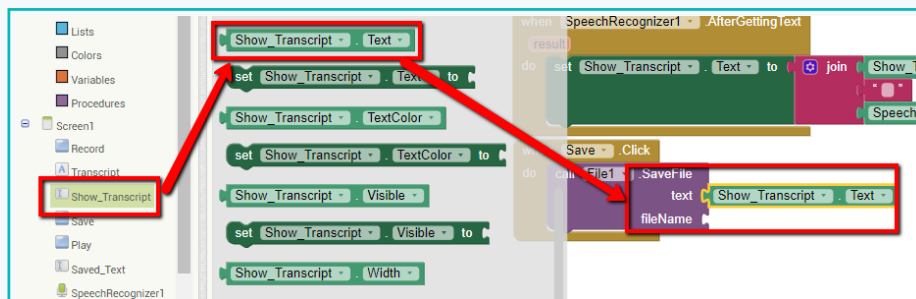


Activity 1

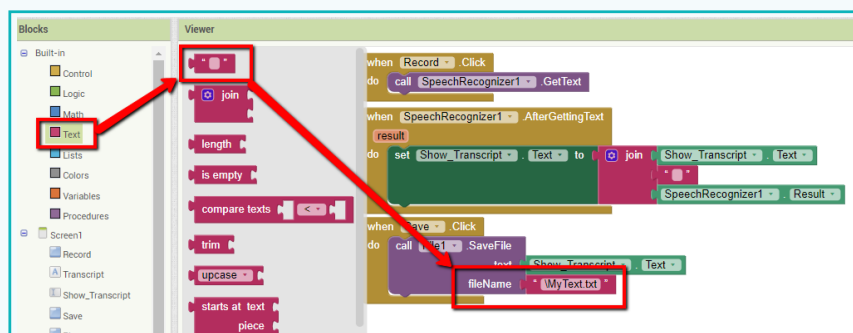
- i. To save the transcript text in a text file, place “when_save.click” from “Save” component on the viewer section. Connect “call_File1.SaveFile” from the “File1” component to “when_Save.Click”.



- j. Connect “Show_Transcript.Text” from the “Show_Transcript” component to “text” in “when_Save.Click”.



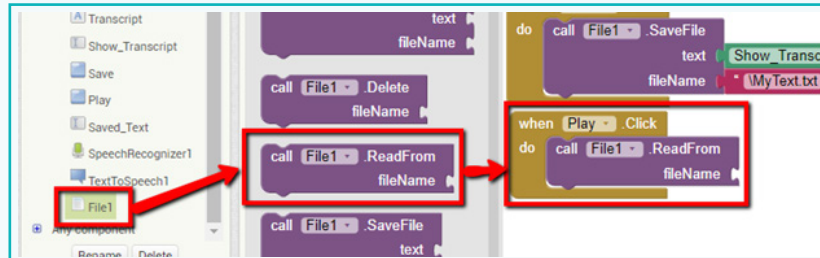
- k. Connect a text constant to “fileName” in the “when_Save.Click” block.



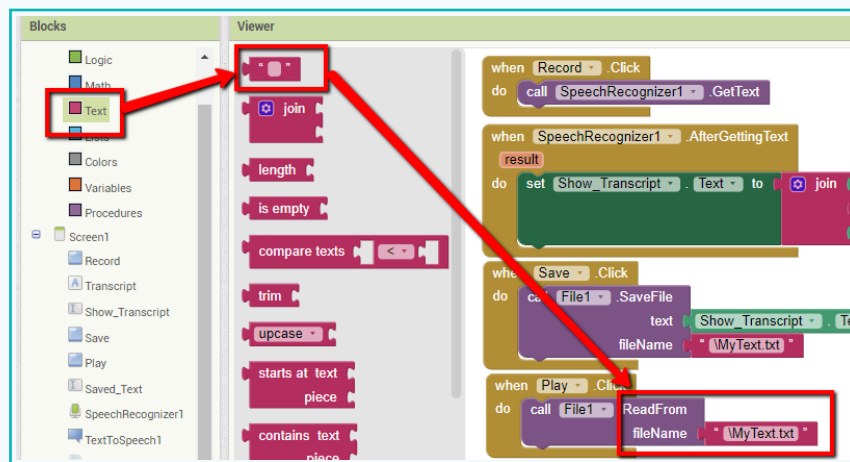
- l. To call text from the text file and display in a text box, place a “when_Play.Click” from the “Play” component on the viewer section.

Activity 1

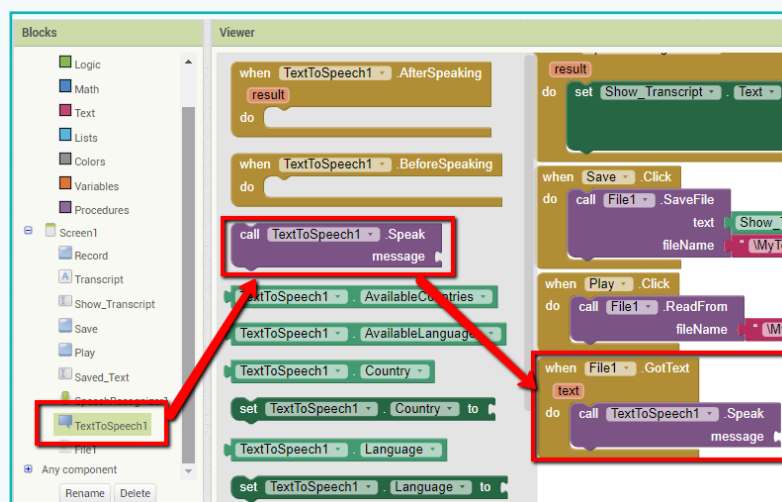
- m. Connect “call_File1.ReadFrom” from the “File1” component to “when_Play.Click”.



- n. Connect a text constant to “fileName” of “call_file1_ReadFrom” block connected to the “when_Play.Click” block.

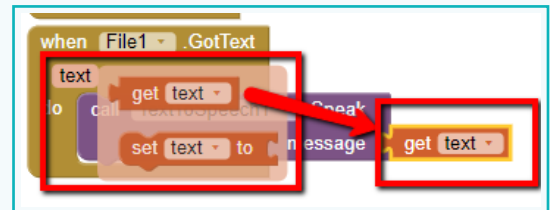


- o. Place a “when_File1.GotText” from the “File1” component onto the viewer section. Connect “call_TextToSpeech1.Speak” from “TextToSpeech1” to “when_File1.GotText”.



Activity 1

- p. Connect “get_text” to “message” of the “call_TextToSpeech1. Speak” block.
- q. To call the text from the text file and display it in a text box, connect “set_Saved_Text.Text.to” from the “saved_text” component under “call_TextToSpeech1.Speak”. Connect “get_text” to it.



Activity 2

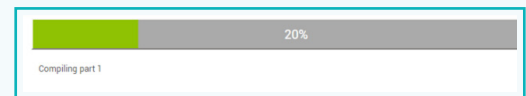
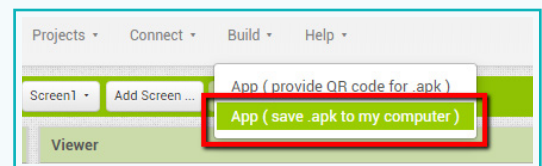
Test the speech recognition application via apk file

There are two ways to test an app.

Emulator - The app is tested on a PC only with the help of an Android emulator. This is useful when we don't have access to a smartphone while developing an app.

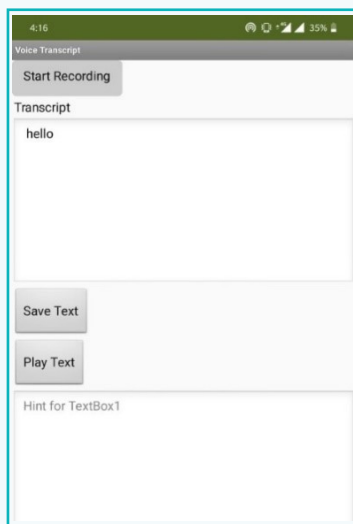
Apk file - An apk file is downloaded on the computer, then copied and installed on a phone via a USB cable.

- 1 Go back to MIT App Inventor on the computer and click on “Build” at the top of designer screen. Click on “App (save .apk to my computer)”.
- 2 Wait for the process to be 100% complete.
- 3 Copy the downloaded application from the “downloads” folder and paste it into the Android phone.
- 4 An option to install the App will show up. Click on “Install”.
- 5 Open the application.
- 6 To take notes of a speech, click “Start Recording” and say “Hello”.

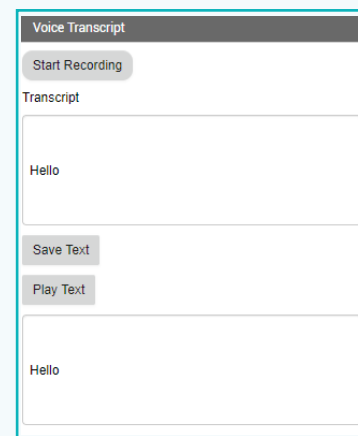


Activity

The app should transcribe the spoken word or sentence and display it on the “Transcript” text box.

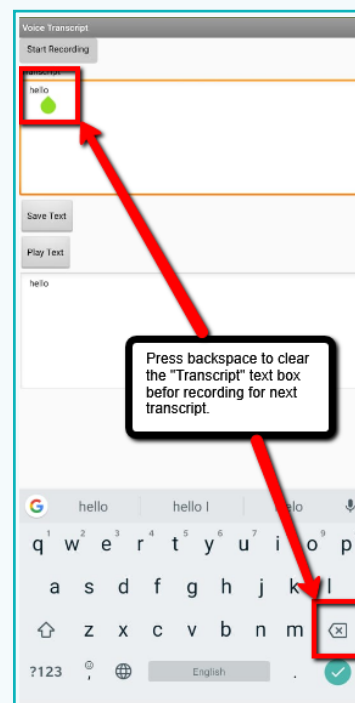


- 7 Click “Save” to save the transcript as a file on the smartphone.
- 8 To read back the transcript, click the “Play Text” button. It should read the contents of the file and convert the text into speech.



Note:

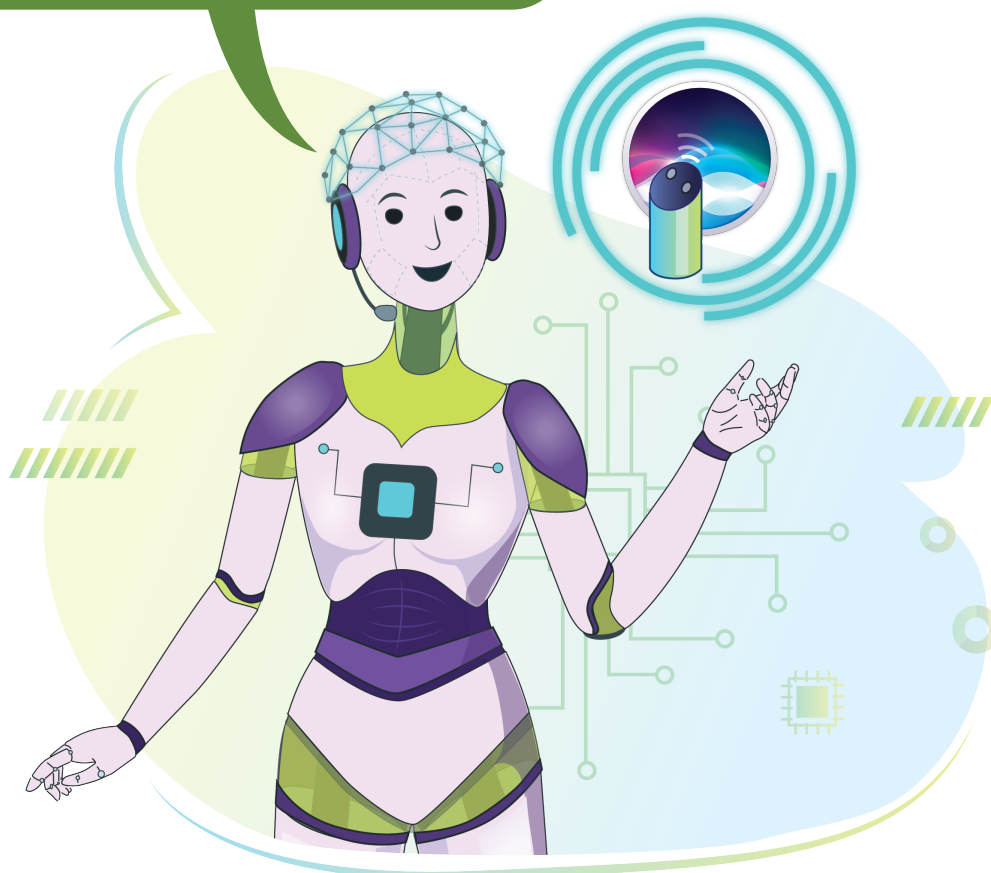
To create a new transcript without saving the previous word/sentence (overwriting it), click on the “Transcript” text box and press backspace to manually clear the text. Click “Start Recording” again to create a new transcript.



Sensors play an integral part in data acquisition. They gather data from around the smartphone and convert it into electrical signals before displaying them to the user in a manner which can be read.

The app we have created will help us transform speech into text and vice versa. The app will use a microphone to record sound and display it in the form of text.

Many devices use the same concept as this. They listen to speech and respond, just like Siri and Alexa.



Tech Flash

- **Data Acquisition** : The process of acquiring data from the world and converting it into digital form is called Data Acquisition.
- **Sensor** : An electronic device that converts physical inputs from the environment into an electronic signal is called a sensor.